Predictive Analytics for Financial Transaction Fraud Detection

Criselvi Lindsey

Bellevue University

DSC 680

Project 1 – Revised White Paper

Professor: Xu Ashton

October 15, 2026

Predictive Analytics for Financial Transaction Fraud Detection

Business Problem and Research Questions

Financial transaction fraud creates financial losses and investigative workload. Predictive analytics can help identify transactions for review, but useful fraud detection must balance detecting fraud with limiting false alerts.

This project evaluates machine-learning classification methods using the BankSim Synthetic Bank Payments Dataset. It asks four questions: (1) How effectively can the models distinguish fraudulent from legitimate transactions? (2) Which transaction characteristics are most useful for identifying fraud? (3) How do the models compare in detecting fraud while minimizing false alerts? and (4) How does class imbalance affect model performance?

Background and Data

Lopez-Rojas and Axelsson (2014) developed BankSim for fraud-detection research using synthetic transactions derived from aggregated transactional patterns. This project uses 594,643 transactions covering approximately six months: 587,443 legitimate and 7,200 fraudulent. Fraud represents 1.21% of the dataset.

The dataset includes transaction sequence, customer, coded age group, gender, customer ZIP code, merchant, merchant ZIP code, transaction category, amount, and fraud status. Table 1 provides the key fields used in the analysis.

Table 1

Key BankSim Variables

Variable	Description	Use in Modeling
step	Transaction sequence across the simulation	Predictor
age	Coded customer age group	Predictor
gender	Coded customer gender	Predictor

merchant	Simulated merchant identifier	Predictor
category	Transaction category	Predictor
amount	Transaction amount	Predictor
fraud	0 = legitimate; 1 = fraudulent	Outcome

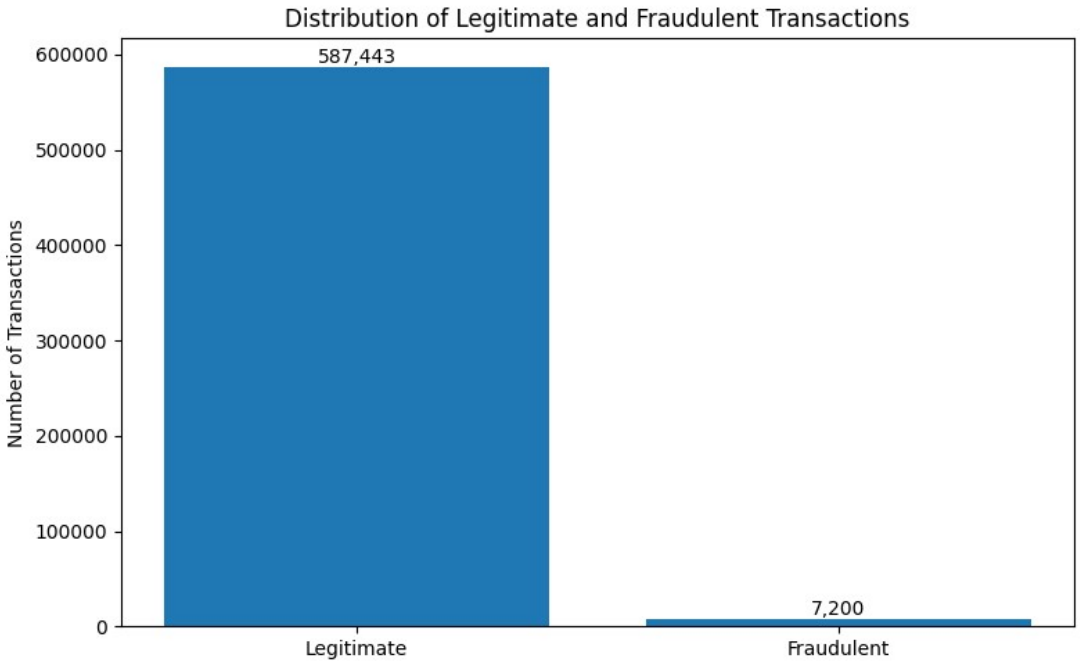
The initial review found no missing values or duplicate records. The analysis excluded zipcodeOri and zipMerchant because each contained only one value. The customer identifier was also excluded to reduce the risk of learning patterns tied to specific simulated customers. The accompanying Python notebook documents the preparation steps.

Exploratory Analysis

The original labels reveal substantial class imbalance: 98.79% of transactions are legitimate and 1.21% are fraudulent.

Figure 1

Distribution of Legitimate and Fraudulent Transactions

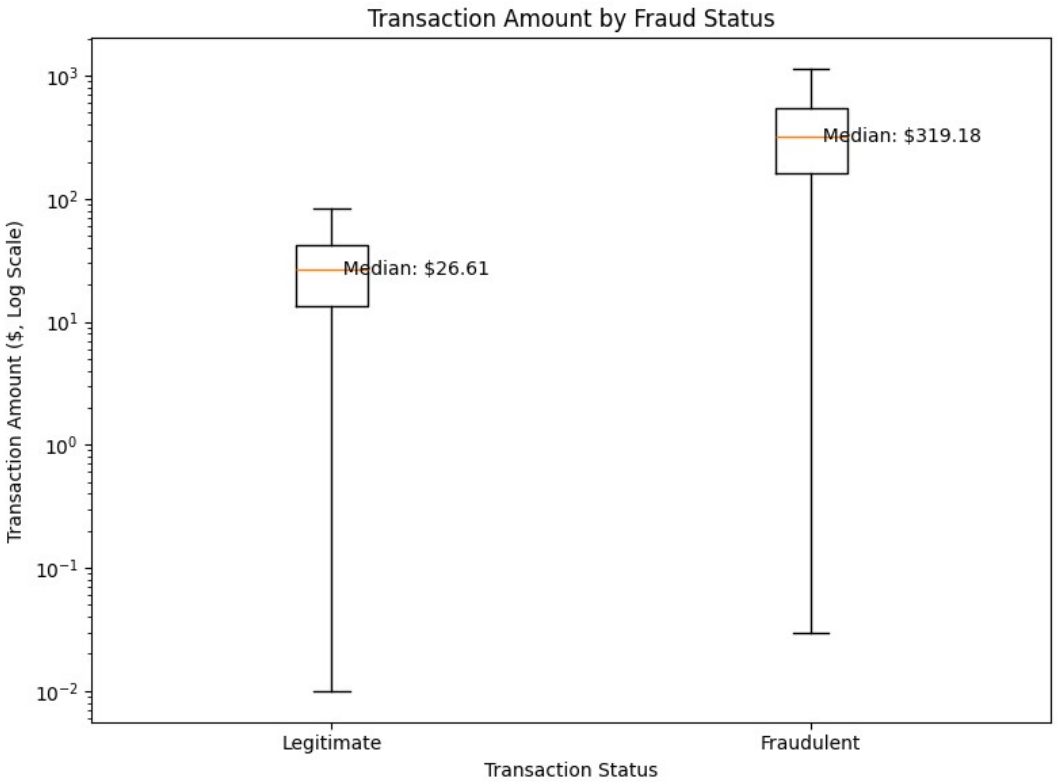


Exploratory analysis showed differences in transaction amount, category, and merchant. The median fraudulent transaction was $319.18 versus $26.61 for legitimate transactions. Figure

2 uses a logarithmic scale to show this difference; zero-dollar transactions remained in the analysis but were omitted from the figure because a log scale cannot display zero.

Figure 2

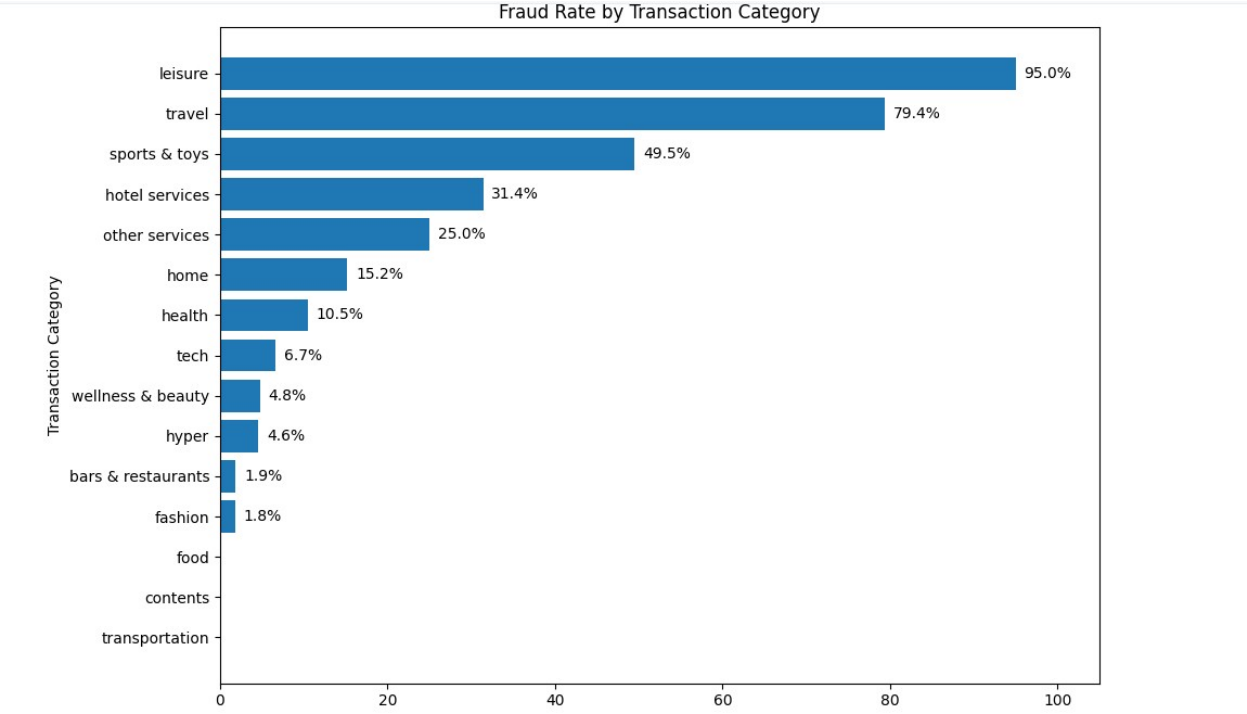
Transaction Amount by Fraud Status



Fraud rates varied considerably by category (Figure 3). Leisure had the highest rate at 94.99%, while sports and toys contained more fraud cases (1,982 versus 474). This distinction shows why both fraud rate and fraud volume matter.

Figure 3

Fraud Rate by Transaction Category



Merchant patterns were also concentrated: M480139044 contained 1,634 fraud cases (23% of all fraud), while M1294758098 had the highest fraud rate (96.34%) but only 184 cases. Age and gender showed smaller differences. Every one of the 180 simulation steps contains exactly 40 fraud cases, so the per-step fraud rate falls mechanically from about 1.65% to 1.07% as transaction volume grows. This regularity reflects BankSim's simulation design rather than fraud behavior, and both the merchant and step patterns are tested in the sensitivity analysis.

Methods

The analysis compares four classifiers, each included for a specific reason. Logistic regression provides an interpretable linear baseline that more complex models must beat. Class-weighted logistic regression keeps the same model and features but up-weights fraud cases, isolating the effect of the most direct remedy for class imbalance. Random forest averages many

decision trees to capture nonlinear relationships and interactions, such as a high amount being suspicious in some categories but not others. Histogram-based gradient boosting builds trees sequentially to correct earlier errors; it is typically among the strongest methods for tabular data and is efficient at this data volume.

The data were divided into 80% training and 20% testing sets using stratified sampling, preserving the 1.21% fraud rate. The test set contained 118,929 transactions, including 1,440 known fraud cases. The test set was reserved for final evaluation: model comparison, tuning, and threshold selection were carried out with cross-validation inside the training set.

Numeric and categorical variables were prepared within modeling pipelines to reduce information leakage. The logistic models standardized numeric features and one-hot encoded nominal features; random forest used one-hot encoding; gradient boosting used ordinal encoding, and an additional experiment compared that approach with one-hot encoding. Sensitivity tests removed merchant, transaction step, and both features, and a time-based validation trained on simulation steps 0–143 and tested on steps 144–179 to assess how strongly the model depended on simulation-specific patterns.

Because fraud is rare, accuracy alone is insufficient. Evaluation therefore emphasizes precision (how often a fraud alert is correct), recall (how much known fraud is detected), and F1 (the balance between precision and recall), along with PR-AUC, false-positive and false-negative rates, and confusion-matrix results.

Validation used five-fold stratified cross-validation of all four models on the full training set (475,714 transactions). Logistic regression was tuned with a grid search over regularization strength and class weighting (eight candidates), and gradient boosting with a randomized search over 20 of 216 combinations of learning rate, number of iterations, tree size, minimum leaf size,

and L2 regularization; both searches used five-fold cross-validation scored by PR-AUC. Decision thresholds for the tuned gradient-boosting model were selected from out-of-fold training predictions, maximizing F1 or, alternatively, keeping recall at or above 0.85, and each was applied once to the test set. Subgroup performance by coded age and gender was evaluated with Wilson 95% confidence intervals and chi-square tests. Analyses were run in Python 3.12.4 with scikit-learn 1.8.0.

Analysis and Results

Table 2

Model Performance on the Test Dataset at the Default 0.50 Threshold

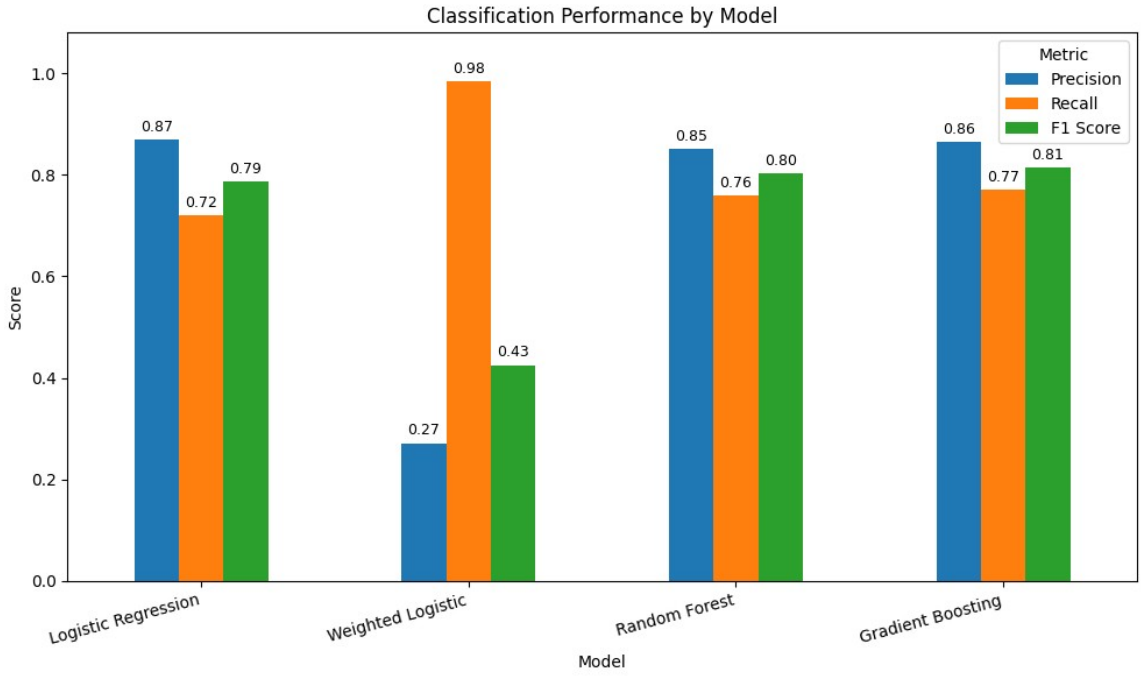
Model	Precision	Recall	F1	PR-AUC	Detected	Missed	False Alerts
Logistic Regression	.870	.719	.788	.883	1,036	404	155
Weighted Logistic	.271	.983	.425	.877	1,416	24	3,804
Random Forest	.851	.759	.803	.872	1,093	347	191
Gradient Boosting	.864	.770	.815	.896	1,109	331	174

The baseline logistic regression detected 1,036 of the 1,440 known fraud cases with 155 false alerts. Balanced class weights increased recall to 98.33%, detecting 1,416 fraud cases, but also produced 3,804 false alerts. These results illustrate the project's central tradeoff between detecting more fraud and creating additional investigative workload.

Random forest achieved 75.90% recall, while gradient boosting achieved 77.01% recall, 86.44% precision, an F1 score of 0.815, and PR-AUC of 0.896. Gradient boosting had the strongest single-split PR-AUC and F1 among the four models. Whether this ranking holds beyond one split is examined through expanded validation below.

Figure 4

Classification Performance by Model

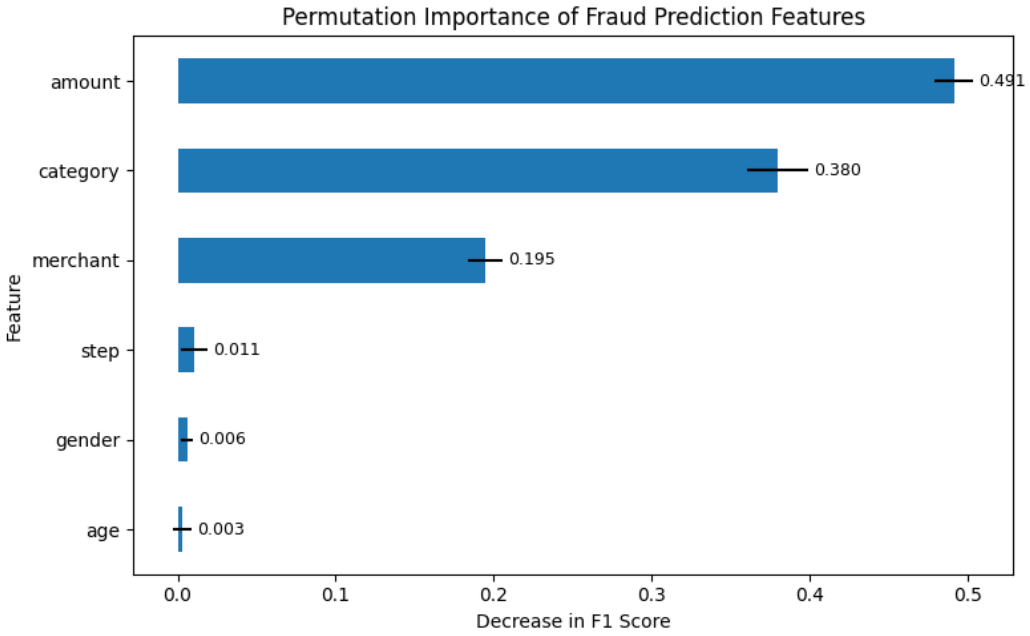


These findings address the first and third research questions by showing that the models identify much of the known fraud while also demonstrating that model choice changes the balance between detection and false alerts. The weighted logistic model especially highlights the operational effect of class imbalance.

Permutation importance for gradient boosting identified amount, category, and merchant as the most influential predictors for F1 performance (Figure 5). Step, age, and gender each reduced F1 by less than 0.02 when shuffled.

Figure 5

Permutation Importance of Fraud Prediction Features



The feature-importance results indicate that transaction amount, category, and merchant are the most useful predictors for this fitted model. These are predictive relationships within BankSim and should not be interpreted as evidence of causation. The negligible importance of age and gender does not by itself guarantee equal performance across those groups, which is evaluated separately below.

Expanded Validation and Sensitivity Analysis

Cross-validation. Across five stratified folds of the training data, gradient boosting had the highest mean PR-AUC (0.8965, SD 0.0087) and F1 (0.8137, SD 0.0086), ahead of logistic regression (PR-AUC 0.8818) and random forest (0.8803) (Table 3, Figure 6). The difference in mean PR-AUC was approximately 0.015–0.016. Fold-to-fold variation was relatively small, although the mean scores alone do not establish that gradient boosting won every fold. Weighted logistic regression again showed very high recall (0.9905) with low precision (0.2738).

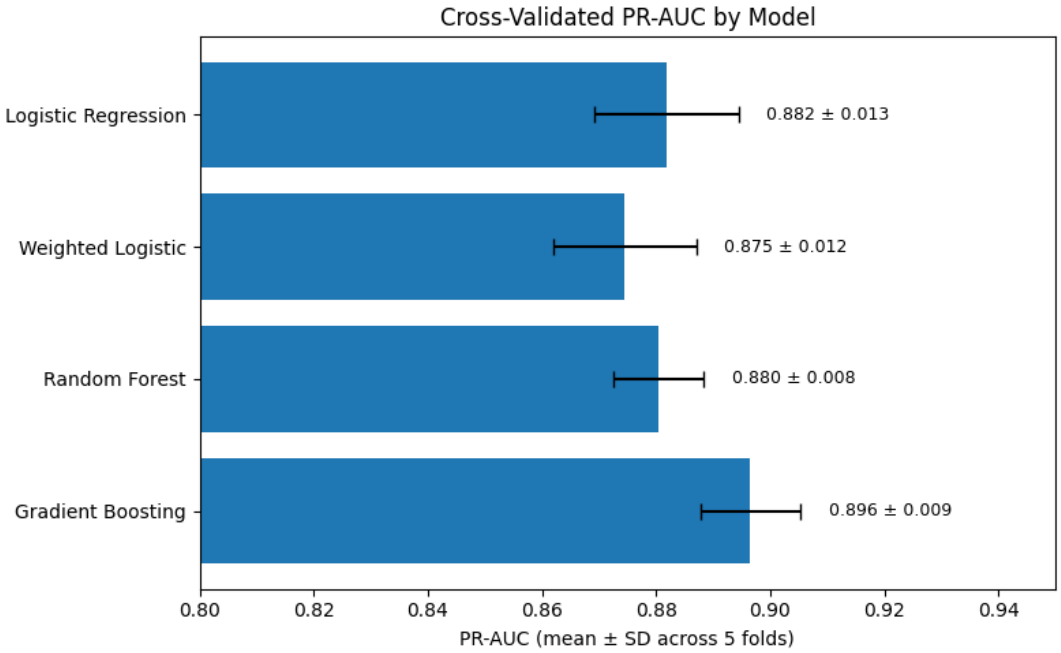
Table 3

Five-Fold Stratified Cross-Validation Results on the Training Set (Mean ± SD)

Model	PR-AUC	Precision	Recall	F1
Logistic Regression	.882 ± .013	.885 ± .016	.723 ± .010	.796 ± .012
Weighted Logistic	.875 ± .012	.274 ± .003	.991 ± .002	.429 ± .004
Random Forest	.880 ± .008	.856 ± .014	.766 ± .006	.809 ± .007
Gradient Boosting	.896 ± .009	.866 ± .021	.768 ± .012	.814 ± .009

Figure 6

Cross-Validated PR-AUC by Model



Hyperparameter tuning. For logistic regression, grid search selected the default regularization (C = 1.0) without class weighting (CV PR-AUC 0.8818). The tuned logistic model's test PR-AUC was 0.8833, matching the baseline. For gradient boosting, the best of 20 sampled configurations used learning rate 0.03, 400 iterations, 15 leaf nodes, minimum leaf size 20, and no L2 penalty. It increased CV PR-AUC from 0.8965 to 0.9040, while the tuned model

achieved test PR-AUC 0.8982. At the default 0.50 threshold, the tuned model detected 1,084 of 1,440 fraud cases with 157 false alerts (precision 0.8735, recall 0.7528, F1 0.8087). Tuning modestly improved ranking performance; threshold selection had a larger practical effect on the balance between fraud detection and false alerts.

Decision-threshold selection. To avoid selecting a threshold on the test set, thresholds were chosen from out-of-fold training predictions. F1 peaked at a threshold of 0.36, while the training threshold selected to target at least 0.85 recall was 0.22 (Figure 7). Applied to the held-out test set (Table 4), the 0.36 threshold detected 1,145 of 1,440 fraud cases with 225 false alerts (precision 0.8358, recall 0.7951, F1 0.8149). The 0.22 threshold detected 75 additional fraud cases, raising recall to 0.8472 but increasing false alerts to 368. The latter fell slightly below the training recall target on the test set. The choice between these operating points depends on the costs of missed fraud and available investigation capacity; the 0.36 threshold is used for the final subgroup analysis.

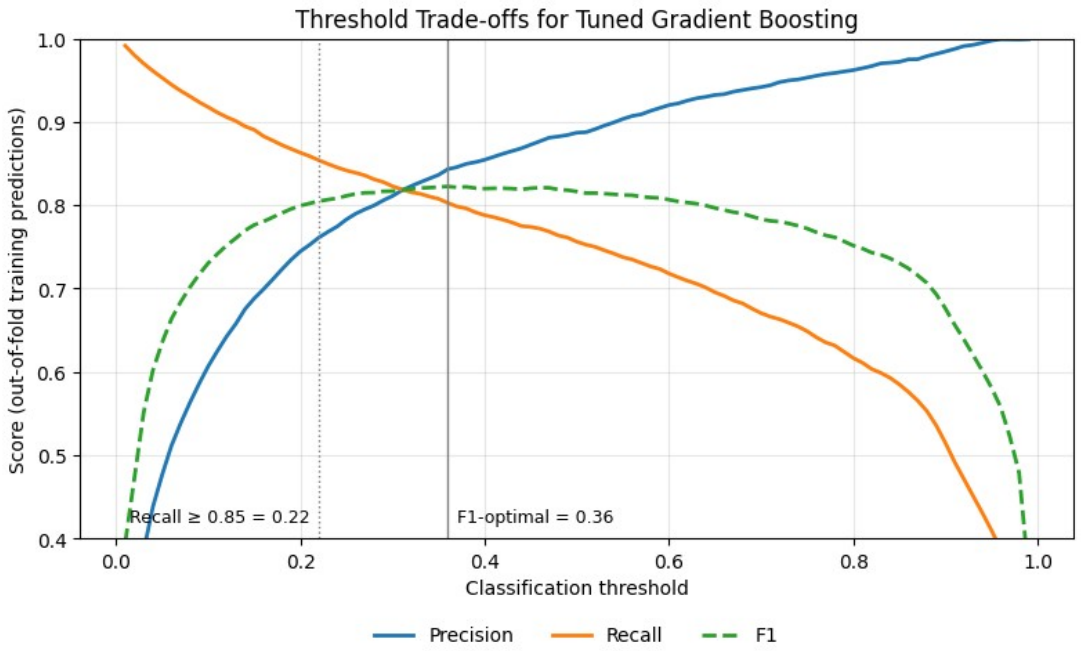
Table 4

Tuned Gradient Boosting on the Test Set at the Default and Out-of-Fold-Selected Thresholds

Operating point	Threshold	Precision	Recall	F1	Detected	Missed	False alerts
Default	0.50	.873	.753	.809	1,084	356	157
F1-optimal	0.36	.836	.795	.815	1,145	295	225
Recall ≥ .85 target	0.22	.768	.847	.806	1,220	220	368

Figure 7

Precision, Recall, and F1 by Threshold (Out-of-Fold Training Predictions)



Feature ablation and time-based validation. On the random split, removing merchant reduced gradient-boosting PR-AUC from 0.8960 to 0.8050 and recall from 0.7701 to 0.6500; removing step had little effect (PR-AUC 0.8952, recall 0.7757). Removing both merchant and step produced PR-AUC 0.8053 (Table 5). Because removing step as a predictor does not remove the fixed 40-fraud-cases-per-step pattern, a separate time-based validation trained on steps 0–143 and tested on steps 144–179. Time-split PR-AUC was 0.9209 with all features and 0.9227 without step, compared with 0.8079 without merchant. These findings suggest limited reliance on step as a predictor but substantial dependence on merchant-specific relationships that might not transfer to real institutions. This does not prove the absence of simulation artifacts or data leakage. Time-split scores are not directly comparable with random-split scores because the fraud rates differ.

Table 5

Gradient-Boosting Feature Ablation Under Random and Time-Based Splits (Threshold 0.50)

Features excluded	Random split PR-AUC	Random split recall	Time split PR-AUC	Time split recall
None	.896	.770	.921	.754
Step	.895	.776	.923	.730
Merchant	.805	.650	.808	.625
Merchant and step	.805	.642	.807	.622

Note. Random split uses the original gradient-boosting settings on the stratified 80/20 split; time split uses the tuned settings, training on steps 0–143 (5,760 fraud cases) and testing on steps 144–179 (1,440 fraud cases).

Categorical encoding. Under the same gradient-boosting settings, one-hot encoding produced test PR-AUC 0.8921, precision 0.8733, and recall 0.7562, compared with PR-AUC 0.8960, precision 0.8644, and recall 0.7701 for ordinal encoding. These modest differences suggest that the artificial ordering imposed by ordinal encoding did not materially drive the results in this comparison. Merchant identity remained influential under both approaches.

Subgroup performance. The final model (tuned gradient boosting at threshold 0.36) was evaluated within each coded age and gender group (Table 6, Figure 8). Fraud recall was similar for gender F (0.7985, 95% CI [0.772, 0.823]) and M (0.7878, [0.749, 0.822]); $\chi^2(1) = 0.16$, $p = .685$. False-positive rates were also similar (0.21% versus 0.17%; $p = .231$). Across age groups 1–5, recall ranged from 0.7677 to 0.8145. Age group 6, described in the dataset documentation as over 65 (Lopez-Rojas, 2017), had lower recall of 0.6889 (95% CI [0.543, 0.805]) versus overall recall of 0.7951. Its confidence interval overlaps the overall recall, and the age-group comparison was not statistically significant, $\chi^2(5) = 4.55$, $p = .473$. With only 45 fraud cases in age group 6, this warrants further investigation rather than a conclusion of unfairness. Groups with fewer than 30 fraud cases (age 0 and U; gender E and U) were excluded from significance tests.

Table 6

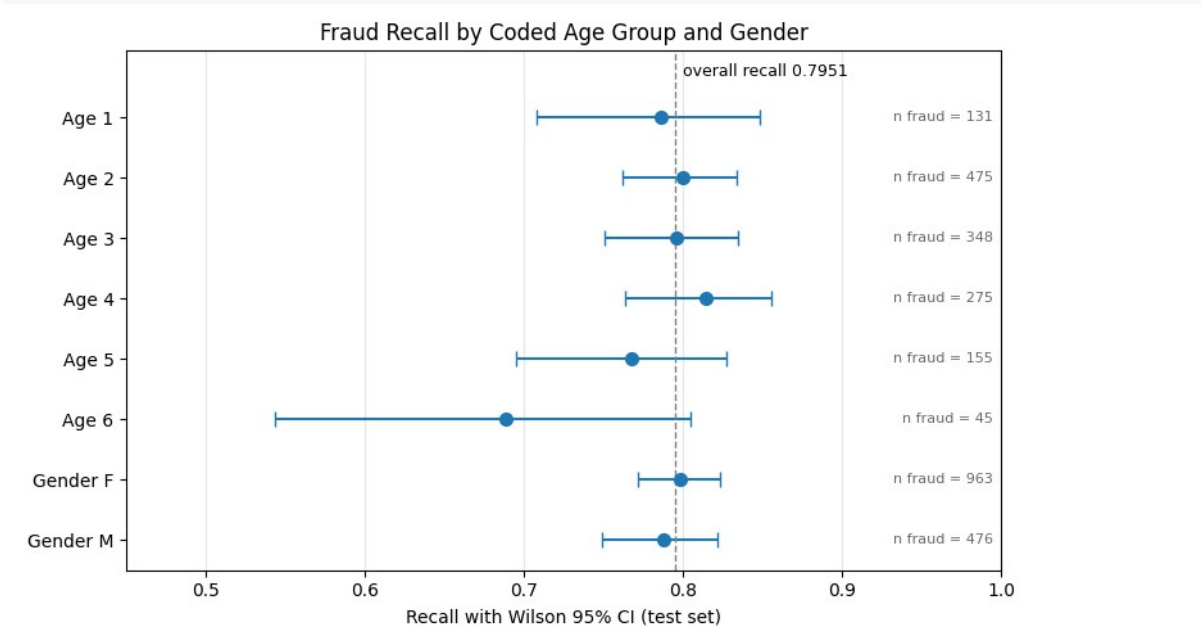
Final-Model Performance by Coded Subgroup on the Test Set (Threshold 0.36)

Group	Test fraud cases	Recall	Recall 95% CI	Precision	False-positive rate
Age 1	131	.786	[.708, .848]	.786	0.25%
Age 2	475	.800	[.762, .834]	.819	0.23%
Age 3	348	.796	[.751, .835]	.839	0.18%
Age 4	275	.815	[.764, .856]	.865	0.16%
Age 5	155	.768	[.695, .827]	.888	0.12%
Age 6	45	.689	[.543, .805]	.816	0.13%
Gender F	963	.799	[.772, .823]	.853	0.21%
Gender M	476	.788	[.749, .822]	.801	0.17%

Note. Confidence intervals are Wilson score intervals. Groups with fewer than 30 test fraud cases (age 0 and U; gender E and U) are omitted; their results appear in the notebook.

Figure 8

Fraud Recall by Coded Age Group and Gender With 95% Confidence Intervals



Conclusion and Recommendations

The analysis indicates that machine-learning models can identify fraud patterns in BankSim, but detection must be assessed alongside false alerts and dependence on simulated features. Gradient boosting performed strongest overall among the four models tested. At the out-of-fold-selected threshold of 0.36, the tuned model detected 1,145 of 1,440 test fraud cases and incorrectly flagged 225 of 117,489 legitimate transactions. Expanded tests showed modest improvements from tuning, substantial trade-offs from threshold choice, material reliance on merchant information, and little sensitivity to the step predictor. Recall was lower in the oldest coded age group, although the overall age-group test was not significant. These are internal synthetic-data results, not estimates of production performance.

For a financial institution, model output should be used as a risk-screening aid rather than an automatic fraud determination. Before deployment, representative institutional data and temporal validation should test merchant dependence, calibration, subgroup performance (particularly for older customers), and operational stability. The operating threshold should be chosen with the out-of-fold procedure used here, applied to institutional data with documented investigation costs and alert capacity, followed by a final untouched-test evaluation and human review of consequential alerts.

Assumptions, Limitations, Challenges, and Future Use

The analysis assumes BankSim labels provide the supervised outcomes and synthetic patterns are useful for controlled model comparison. They may not represent actual customer behavior. Feature ablation confirmed that merchant information materially influenced performance (random-split PR-AUC 0.8960 with all features versus 0.8050 without merchant, and a similar loss under the time split), whereas removing step had little effect. This is a model-

dependence finding, not proof of target leakage. The fixed 40 fraud cases per step remain a simulation artifact that the time-based validation cannot fully rule out. Ordinal encoding imposes numerical order on nominal categories, but one-hot encoding produced similar results.

Class imbalance was addressed with stratified splitting and folds, PR-AUC and class-specific metrics, a class-weighted comparison, and threshold selection. The hyperparameter search for gradient boosting sampled 20 of 216 configurations rather than the full grid, so a better configuration may exist, although the small spread among the top configurations suggests limited remaining gains. The time-based validation used a single cut-off within one six-month simulation. Subgroup results rely on synthetic coded categories, and the age-group-6 estimate rests on 45 test fraud cases. Future work should evaluate rolling time-based validation, external samples, cost-weighted thresholds, probability calibration, and fairness under actual deployment conditions.

Ethical Assessment

Identify. Key risks are missed fraud, unnecessary customer alerts, unequal subgroup performance, dependence on simulation-specific patterns, performance drift as fraud tactics change, privacy, and inappropriate reliance on automated scores. Although BankSim contains synthetic transactions, real financial data would require access controls and data minimization. Age and gender were evaluated empirically using their coded categories (Table 6).

Assess. Table 7 rates each risk for likelihood and impact on a three-point scale (1 = low, 2 = medium, 3 = high) and multiplies the two into a risk score from 1 to 9. The ratings are analyst judgments for a hypothetical deployment, informed by the evidence shown; they are not probabilities measured from BankSim. The two highest-scoring risks, merchant-dependent simulation patterns and missed fraud, are both demonstrated directly by this analysis. Subgroup

disparity is rated medium likelihood because the oldest age group showed lower recall even though the overall test was not significant.

Table 7

Ethical Risk Register With Likelihood, Impact, and Risk Scores

Risk	Likelihood	Impact	Score	Evidence from this analysis
Merchant-specific simulation patterns do not transfer	High (3)	High (3)	9	Removing merchant cut PR-AUC by 0.091 (random split) and 0.113 (time split)
Missed fraud (false negatives)	High (3)	High (3)	9	Final model missed 295 of 1,440 test fraud cases (20.5%)
Unequal performance across age groups	Medium (2)	High (3)	6	Age 6 recall 0.689 (CI 0.543–0.805) vs. 0.795 overall; p = .473
Customer disruption from false alerts	High (3)	Medium (2)	6	225 false alerts at 0.36; 368 at 0.22
Overreliance on automated scores	Medium (2)	High (3)	6	About 1 in 6 alerts is a legitimate transaction (precision 0.836)
Performance drift as fraud tactics change	Medium (2)	High (3)	6	Time-split PR-AUC 0.921; real fraud patterns may change
Privacy breach or unauthorized access	Low (1)	High (3)	3	Not testable with synthetic data; real data are highly sensitive

Mitigate. Validate on representative institutional data and repeat the merchant ablation before deployment. Select the threshold with the out-of-fold procedure using documented investigation costs and alert capacity, and route consequential cases to trained human reviewers with a documented appeal and override process. Monitor fraud recall, false-positive rates, alert volume, and calibration overall and by subgroup with confidence intervals, and conduct a larger-sample audit of older customers. Use drift monitoring and scheduled retraining. Apply least-privilege access, retention limits, audit logging, and privacy review to real transaction data.

Code and Reproducibility

The accompanying executed Python notebook documents each step in Markdown, including the rationale for every model, and follows each major output with a written interpretation. It covers loading, data-quality checks, exploratory analysis, stratified splitting,

pipelines, the four-model comparison, permutation importance, five-fold cross-validation of all models, hyperparameter tuning, out-of-fold threshold selection, feature ablation, time-based validation, the encoding comparison, subgroup analysis with confidence intervals, and the ethical risk register. All code cells were executed in order without errors using Python 3.12.4, pandas 2.2.3, and scikit-learn 1.8.0 (versions are printed in the notebook), with random state 42. Rerunning requires the BankSim dataset; tree-model results may shift slightly under other scikit-learn versions.

References

Lopez-Rojas, E. A. (2017). Synthetic data from a financial payment system [Data set]. Kaggle. https://www.kaggle.com/datasets/ealaxi/banksim1

Lopez-Rojas, E. A., & Axelsson, S. (2014). BankSim: A bank payments simulator for fraud detection research. Proceedings of the 26th European Modeling & Simulation Symposium, 144–152.

Appendix A
Supporting Documentation

Supporting documentation is provided in the accompanying executed Python notebook, including the original models and the expanded analyses: five-fold cross-validation of all four models, logistic-regression and gradient-boosting tuning, out-of-fold threshold selection, merchant/step feature ablation, time-based validation, encoding comparison, age and gender subgroup analysis with confidence intervals and significance tests, and the scored ethical risk register.